

Problem 1:

Let R be a ring. Prove that the following conditions are equivalent:

- (a) R is a local ring.
- (b) $\text{Spec}(R)$ has a unique closed point.
- (c) There exists a proper ideal $\mathfrak{m} \subsetneq R$ s.t. every element of $R/\mathfrak{m}$ is invertible.
- (d) R is nonzero and for any element $x \in R$, either x or $1 - x$ is invertible.

Claim

There is a bijection between

$$A = \{\text{Closed points of } \text{Spec}(R)\} \leftrightarrow B = \{\text{Maximal ideals in } R\}$$

Proof.

Define $\varphi : A \rightarrow B$ s.t. $\varphi(\{\mathfrak{m}\}) = \mathfrak{m}$.

Well-Defined:

Let $\mathfrak{m}$ be a closed point in $\text{Spec}(R)$ then shown in class

$$\{\mathfrak{m}\} = \overline{\{\mathfrak{m}\}} = V(\mathfrak{m}) = \{\mathfrak{q} \in \text{Spec}(R) \mid \mathfrak{m} \subseteq \mathfrak{q}\}$$

This implies there is no prime ideal that contains $\mathfrak{m}$.

Assume for contradiction that $\mathfrak{m}$ is not maximal then there is an ideal I s.t. $\mathfrak{m} \subsetneq I \subsetneq R$. Since every ideal is contained in a maximal ideal. This implies that $I \subset \mathfrak{M}$ where $\mathfrak{M}$ is maximal. However, this is a contradiction since $\mathfrak{M}$ is prime and no prime ideal contains $\mathfrak{m}$. Thus, $\mathfrak{m}$ is maximal and well-defined.

Injective:

Let $\{\mathfrak{m}\}, \{\mathfrak{n}\} \in A$ s.t. $\varphi(\{\mathfrak{m}\}) = \varphi(\{\mathfrak{n}\})$.

This implies that $\mathfrak{m} = \mathfrak{n}$. Thus, $\{\mathfrak{m}\} = \{\mathfrak{n}\}$.

Surjective:

Let $\mathfrak{m}$ be a maximal ideal in R . Consider $\{\mathfrak{m}\}$. Show in class.

$$\overline{\{\mathfrak{m}\}} = V(\mathfrak{m}) = \{\mathfrak{q} \in \text{Spec}(R) \mid \mathfrak{m} \subseteq \mathfrak{q}\}$$

Since $\mathfrak{m}$ is maximal this implies that $\overline{\{\mathfrak{m}\}} = \{\mathfrak{m}\}$. Thus, $\varphi(\{\mathfrak{m}\}) = \mathfrak{m}$.

Therefore, φ is a bijection. ■

Proof.

(a) $\iff$ (b):

By the claim above since closed points of $\text{Spec}(R)$ is in bijection with Maximal ideals in R . This implies that R has a unique maximal ideal if and only if $\text{Spec}(R)$ has a unique closed point.

(a) $\implies$ (c):

Assume (a). Let $\mathfrak{m}$ be the unique maximal ideal of R . Let $x \in R \setminus \mathfrak{m}$.

Consider (x) . Since there is a unique maximal ideal this implies that (x) is not contained in any maximal ideal thus $(x) = R$. Therefore, there exists $r \in R$ s.t. $xr = 1$.

Thus, all elements in $R \setminus \mathfrak{m}$ are invertible.

(c) $\implies$ (d):

Assume (c). Let $\mathfrak{m}$ be a proper ideal s.t. every element of $R \setminus \mathfrak{m}$ is invertible.

Let $x \in R$.

If $x \in R \setminus \mathfrak{m}$ then x is invertible.

Let $ab \in \mathfrak{m}$ for $a, b \in R$. Assume $a \notin \mathfrak{m}$.

This implies that a is invertible.

Thus, $a^{-1}ab = b \in \mathfrak{m}$.

Therefore, $\mathfrak{m}$ is prime since it is proper by assumption.

If $x \in \mathfrak{m}$.

Consider $1 - x$: $1 - x \notin \mathfrak{m}$ since if it was then $(1 - x) + x = 1 \in \mathfrak{m}$ which is a contradiction since $\mathfrak{m}$ is prime.

Thus, by assumption $1 - x \in R \setminus \mathfrak{m}$ thus invertible.

(d) $\implies$ (a):

Assume (d). Let M be the set of nonunits of R .

Let $a, b \in M$ and assume $a + b$ is a unit.

Then $(a + b)(a + b)^{-1} = 1 \implies a(a + b)^{-1} + b(a + b)^{-1} = 1 \implies a(a + b)^{-1} = 1 - b(a + b)^{-1}$.

By assumption either $b(a + b)^{-1}$ or $a(a + b)^{-1}$ is invertible.

If $b(a + b)^{-1}$ is invertible then $b(a + b)^{-1}r = 1$ for some $r \in R$. This implies b is invertible which is a contradiction.

This is the same for if $a(a + b)^{-1}$ is invertible. Thus, $a + b \in M$.

Let $a \in M$ and $r \in R$. ar is not a unit since if it was then $ar \cdot (ar)^{-1} = 1 \implies a$ is a unit which is a contradiction.

Thus, M is an ideal.

M is proper since 1 is a unit.

Thus, M is contained in a maximal ideal $\mathfrak{M}$.

Since there are no units in $\mathfrak{M}$ this implies that for $a \in \mathfrak{M}$ $1 - a \in \mathfrak{M}$ is a unit.

Assume that $\mathfrak{M}$ is not unique. This implies there is another maximal ideal $\mathfrak{N}$ s.t. $\mathfrak{N} \cap R/\mathfrak{M} \neq \emptyset$. This implies that $\mathfrak{N}$ has a unit since all non units are contained in $\mathfrak{M}$. Thus, $\mathfrak{M}$ is a unique maximal ideal. Therefore, R is a local ring. ■

Problem 2:**(a)**

Let R be a ring, $I \subseteq R$ be an ideal, and $S \subset R$ be a multiplicative system. Let $S^{-1}I \subset S^{-1}R$ denote the ideal obtained by the extension of I along the natural ring homomorphism $R \rightarrow S^{-1}R$, i.e. $S^{-1}I$ is ideal generated by the image of I . Let $\pi : R \rightarrow R/I$ be the quotient map and $T = \pi(S) \subset R/I$ be the image of S , which is easily seen to be a multiplicative system. Prove that there is a ring isomorphism

$$T^{-1}(R/I) \cong S^{-1}R/S^{-1}I.$$

Proof.

Let $f : R/I \rightarrow T^{-1}(R/I)$ and $i : R \rightarrow S^{-1}R$ be the standard localization maps. Let $s \in S$. Then $\pi(s) \in T \implies f(\pi(s))$ is a unit. Therefore, by the universal property, there exists a unique map $h : S^{-1}R \rightarrow T^{-1}(R/I)$ that makes the diagram commute.

$$\begin{array}{ccccc} R & \xrightarrow{\pi} & R/I & \xrightarrow{f} & T^{-1}(R/I) \\ & \searrow i & & \nearrow h & \\ & & S^{-1}R & & \end{array}$$

Surjective:

Let $r/t \in T^{-1}(R/I)$ then $r \in R/I$ and $t \in T$.

Since π is Surjective let $\pi(r') = r$ and $\pi(t') = t$ for $r' \in R$ and $t' \in S$.

Thus, $h(r'/t') = \pi(r')/\pi(t') = r/t$.

Kernel:

Let $j/s \in S^{-1}/I$.

$h(j/s) = \pi(j)/\pi(s)$. Since $j \in I \implies \pi(j) = 0$. Thus, $h(j/s) = 0$. Thus, $S^{-1}/I \subset \ker(h)$.

Let $x/s \in \ker(h)$

$h(x/s) = \pi(x)/\pi(s) = 0$. Since $\pi(s)$ is invertible. $\pi(x)/\pi(s) \cdot \pi(s)/1 = \pi(x)/1 = 0$. Thus, there exists a $\pi(t) \in T^{-1}$ s.t. $t \in S$ and $\pi(t) \cdot \pi(x) = 0$.

Thus, $\pi(tx) = 0 \implies tx \in I$.

Thus, $x/s = tx/ts$. Since $tx \in I$ this implies $tx/ts \in S^{-1}I$ thus $x/s \in S^{-1}I$.

Therefore, $\ker(h) = S^{-1}I$.

Thus, by first isomorphism theorem we have that $S^{-1}R/S^{-1}I \cong T^{-1}(R/I)$ ■

(b)

Let R be a ring and $\mathfrak{p} \in \text{Spec}(R)$. Explain why (a) implies there is an isomorphism

$$\kappa(\mathfrak{p}) \cong R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}},$$

where recall that $\kappa(\mathfrak{p}) = \text{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ and $\mathfrak{p}R_{\mathfrak{p}}$ is the maximal ideal of the local ring $R_{\mathfrak{p}}$.

Proof.

Let $\pi : R \rightarrow R/\mathfrak{p}$ be the standard quotient map.

Consider $\pi(R \setminus \mathfrak{p})$

Let $x \in \pi(R \setminus \mathfrak{p})$. Thus, there exists a $x' \in R \setminus \mathfrak{p}$ s.t. $\pi(x') = x$. Since $x' \notin \mathfrak{p} \implies \pi(x') \neq 0$. Thus, $x \in (R/\mathfrak{p} \setminus \{0\})$.

Let $x \in R/\mathfrak{p} \setminus \{0\}$. Since π is surjective this implies there exists a $x' \in R$ s.t. $\pi(x') = x$. Since $x \neq 0 \implies x' \in R \setminus \mathfrak{p}$. Thus, $x \in \pi(R \setminus \mathfrak{p})$.

Thus, $\pi(R \setminus \mathfrak{p}) = R/\mathfrak{p} \setminus \{0\}$.

Since $R/\mathfrak{p}$ is an integral domain this implies that $\kappa(\mathfrak{p}) = \text{Frac}(R/\mathfrak{p}) = (R/\mathfrak{p} \setminus \{0\})^{-1}R/\mathfrak{p}$.

Therefore, by (a) we have that $\kappa(\mathfrak{p}) = \text{Frac}(R/\mathfrak{p}) = (R/\mathfrak{p} \setminus \{0\})^{-1}R/\mathfrak{p} = \pi(R \setminus \mathfrak{p})^{-1}(R/\mathfrak{p}) \cong R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$. ■

Problem 3:**(a)***Proof.*

Let $F : \mathcal{I} \rightarrow \text{Ring}$ be a functor.

Let $R_i := F(i)$ and $f_{ij} := F(\phi_{ij})$.

Define $L := \sqcup_{i \in I} R_i / \sim$

where for $x_i \in R_i$ and $x_j \in R_j$ and $x_i \sim x_j$ if and only iff there exists a $k \in I$ s.t. $k \geq i, j$ and $f_{ik}(x_i) = f_{jk}(x_j)$.

Equivalence Relation:

Reflexive: Since $f_{ij}(x_i) = f_{ij}(x_i)$ for any $j \geq i$, $x_i \sim x_i$.

Symmetric: Assume $x_i \sim x_j$. Then there exists a $k \geq i, j$ s.t. $f_{ik}(x_i) = f_{jk}(x_j) \implies f_{jk}(x_j) = f_{ik}(x_i) \implies x_j \sim x_i$.

Transitive: Assume $x_i \sim x_j$ and $x_j \sim x_k$. Thus, there exists $s \geq i, j$ and $t \geq j, k$ s.t. $f_{is}(x_i) = f_{js}(x_j)$ and $f_{jt}(x_j) = f_{kt}(x_k)$. Let $m \geq s, t$.

Thus, $f_{im}(x_i) = f_{sm}(f_{is}(x_i)) = f_{sm}(f_{js}(x_j)) = f_{tm}(f_{jt}(x_j)) = f_{tm}(f_{kt}(x_k)) = f_{km}(x_k)$.

Therefore, $x_i \sim x_k$.

Thus, $\sim$ is an equivalence relation.

Ring:

$1_L := [1_{R_i}]$ and $0_L := [0_{R_i}]$ for $i \in I$ since $f_{ij}(1) = 1$ and $f_{ij}(0) = 0$ for all $j \geq i$ for $i, j \in I$ since f_{ij} are ring homomorphisms.

Addition + Multiplication:

$$[x_i] * [x_j] = [f_{ik}(x_i) * f_{jk}(x_j)]$$

where $k \geq i, j$ and $*$ is multiplication or addition.

Thus, L has a natural ring structure.

Well-Defined:

Let $x_i \in R_i$ and $x_j \in R_j$ and $k, k' \geq i, j$.

$$[x_i] * [x_j] = [f_{ik}(x_i) * f_{jk}(x_j)] \quad [x_i] * [x_j] = [f_{ik'}(x_i) * f_{jk'}(x_j)]$$

Let $m \geq k, k'$. Thus,

$$[f_{ik}(x_i) * f_{jk}(x_j)] = [f_{im}(x_i) * f_{jm}(x_j)] = [f_{ik'}(x_i) * f_{jk'}(x_j)]$$

Since $f_{km}(f_{ik}(x_i) * f_{jk}(x_j)) = f_{mi}(x_i) * f_{mj}(x_j) = f_{k'm}(f_{ik'}(x_i) * f_{jk'}(x_j))$

If $[x_i] = [x'_i]$ and $[x_j] = [x'_j]$. Let $k \geq i, j$ and $k' \geq i', j'$

Let $m \geq k, k'$ then $f_{mi}(x_i) = f_{mi'}(x_{i'})$ and $f_{mj}(x_j) = f_{mj'}(x_{j'})$. Thus, $*$ agrees in R_m , so addition and multiplication are well-defined.

Colimit:

Let R be another ring s.t. it has maps $\psi_i : R_i \rightarrow R$ s.t. $\psi_j \circ f_{ij} = \psi_i$.

Let $\psi : L \rightarrow R$ s.t. $\psi([x_i]) = \psi_i(x_i)$.

Assume $[x_i] = [x_j]$. Thus, there implies $k \geq i, j$ s.t. $f_{ik}(x_i) = f_{jk}(x_j)$.

Thus, $\psi([x_i]) = \psi_i(x_i) = \psi_k(f_{ik}(x_i)) = \psi_k(f_{jk}(x_j)) = \psi_j(x_j) = \psi([x_j])$.

Also, ψ is a homomorphism since it is defined by ψ_i and these are homomorphism.

This is clearly unique. Thus, L is the colimit of F . ■

(b)

Proof.

Define

$$\mathfrak{m} = \{[f] \in \mathcal{O}_p \mid f(p) = 0\}$$

This is an ideal since if $f(p) = 0 = g(p)$ then $f + g(p) = 0$. Also, if $f(p) = 0$ and g is any germ then $f \cdot g(p) = 0$.

It is proper since $[1] \notin \mathfrak{m}$ and $[1]$ never vanishes.

Thus, all germs in $\mathcal{O}_p \setminus \mathfrak{m}$ are units since they do not vanish on p , so you can divide with them at p .

Thus, by problem one $\mathfrak{m}$ is a proper ideal s.t. $\mathcal{O}_p \setminus \mathfrak{m}$ are all units implies $\mathcal{O}_p$ is a local ring. ■

(c)

Proof.

If $f \mid f'$ with $f' = fg$ for $g \in R$, then the universal property of localization gives a unique map

$$\varphi_{f,f'} : R_f \rightarrow R_{f'}$$

s.t. $\varphi(r/f^n) = rg^n/(f')^n$.

For each $f \in S$ define $\psi_f : R_f \rightarrow S^{-1}R$, s.t. $\psi_f(r/f^n) = r/f^n$. If $f \mid f'$ s.t. $f' = fg$ then

$$\psi_{f'}(\varphi_{f,f'}(r/f^n)) = \psi_{f'}(rg^n/(f')^n) = \frac{rg^n}{(f')^n} = \frac{r}{f^n} = \psi_f(r/f^n)$$

Therefore, by the universal property of the colimit, there exists a unique ring map,

$$\psi : \operatorname{colim}_{f \in S} R_f \rightarrow S^{-1}R$$

with $\psi \circ g_f = \psi_f$ s.t. $g_f : R_f \rightarrow \operatorname{colim}_{f \in S} R_f$ and g_f is the family of morphism associated with the colimit.

Define: $\alpha : R \rightarrow \operatorname{colim}_{f \in S} R_f$ s.t. $\alpha = g_1 \circ \eta$ s.t. $\eta : R \rightarrow R_1 \cong R$.

Let $s \in S$ then $\alpha(s) = g_s(\varphi_{1,s}(s)) = g_s(s)$ which is invertible in R_s thus invertible in the colimit. Thus, by the universal property of localization there exists a unique ring map

$$h : S^{-1}R \rightarrow \operatorname{colim}_{f \in S} R_f$$

with $h \circ v = \alpha$ where $v : R \rightarrow S^{-1}R$ is the localization map.

This is well-defined since if $r/s = r'/s'$ in $S^{-1}R$, there exists $t \in S$ with $t(s'r - sr') = 0$ then in $R_{ss't}$,

$$\varphi_{s,ss't}(r/s) = \frac{rs't}{ss't} = \frac{r'st}{ss't} = \varphi_{s',ss't}(r'/s')$$

Inverses:

Let $r/s \in S^{-1}R$,

$$\psi \circ h(r/s) = \psi(g_s(r/s)) = \psi_s(r/s) = r/s$$

Let $x \in \operatorname{colim}_{f \in S} R_f$ then there exists a $f \in S$ and $r \in R$ s.t. $g_f(r/f^n) = x$.

$$\begin{aligned} h \circ \psi(x) &= h \circ \psi(g_f(x)) = h(\psi_f(x)) = h(r/f^n) = \alpha(r)\alpha(f)^{-n} \\ &= g_{f^n}(r/1)g_{f^n}(1/f^n) = g_{f^n}(r/f^n) = g_f(\varphi_{f,f^n}(r/f^n)) = g_f(r/f^n) = x \end{aligned}$$

Therefore, $S^{-1}R \cong \operatorname{colim}_{f \in S} R_f$ ■

Problem 4:

A topological space X is called *irreducible* if it is nonempty and for any expression of X as a union $X = Z_1 \cup Z_2$ where $Z_1, Z_2 \subset X$ is closed, we have $X = Z_1$ or $X = Z_2$. Consider the ring

$$R = \mathbb{C}[x, y, z, w] / ((x + 2)^5, xyzw + 3xyz + 2yzw + 6yz, (y + z)^4)$$

Determine, with proof, whether $\text{Spec}(R)$ is irreducible.

Claim

$\text{Spec}(R/I)$ is irreducible if and only if $\sqrt{I}$ is prime.

Proof.

Proven in class $\text{Spec}(R/I)$ is irreducible if and only if the nilradical of R/I is prime.

But the nilradical of R/I is $\sqrt{I}/I$, and $\sqrt{I}/I$ is prime if and only if $\sqrt{I}$ is prime.

Thus, combining both implications we get $\text{Spec}(R/I)$ is irreducible if and only if $\sqrt{I}$ is prime. ■

Proof.

Do change of variables where

- $u = x + 1$
- $y = y$
- $t = w + 3$
- $v = y + z$

Thus, $R \cong \mathbb{C}[u, y, v, t] / (u^5, v^4, uy(v - y)t)$.

Shown in class $\text{Spec}(R)$ is irreducible if and only iff the nilradical of R is prime.

Consider $I = (u^5, v^4, uy(v - y)t)$

Since $u^5 \in I$ and $v^4 \in I$ this implies that $(u, v) \subseteq \sqrt{I}$.

Also, $I \subseteq (u, v)$ since $u^5 \in (u, v)$, $v^4 \in (u, v)$, $uy(v - y)t \in (u) \subseteq (u, v)$.

Thus, $\sqrt{I} \subseteq \sqrt{(u, v)}$, but since $\mathbb{C}[u, y, v, t] / (u, v) \cong \mathbb{C}[y, t]$ is an integral domain. This implies (u, v) is prime thus $\sqrt{(u, v)} = (u, v)$.

Therefore, $\sqrt{I} = (u, v)$. This implies that $\text{Spec}(R)$ is irreducible. ■

Problem 5:

Let X be a topological space. An *irreducible component* Z of X is a maximal irreducible subset $Z \subset X$. In other words, Z is irreducible and if $Z' \subset X$ is another irreducible closed subset s.t. $Z \subset Z'$, then $Z = Z'$.

(a)

Prove that if $Z \subset X$ is an irreducible component of X , then Z is closed.

Proof.

Assume Z is an irreducible component. Shown in class since Z is irreducible this implies that $\overline{Z}$ is irreducible. Since $Z \subset \overline{Z}$, this implies $Z = \overline{Z}$. Thus, Z is closed. ■

(b)

Prove that if $Z \subset X$ is an irreducible subset of X , then there exists an irreducible component $W \subset X$ s.t. $Z \subset W$

Proof.

Consider $\Sigma = \{Y \subset X \mid Z \subset Y \text{ and } Y \text{ is irreducible}\}$, and order it by inclusion.

If $\{Y_i\}_{i \in I} \subseteq \Sigma$ is a chain, then $Y = \bigcup_{i \in I} Y_i$ is irreducible.

Let U, V be nonempty open in the subspace Y . Then let $x \in U$ and $y \in V$. There exists $i, j \in I$ with $x \in Y_i$ and $y \in Y_j$. Since $\{Y_i\}_{i \in I}$ is a chain. WLOG let $Y_i \subset Y_j$. Then $U \cap Y_j, V \cap Y_j$ are open in the subspace topology on Y_j . Since Y_j is irreducible this implies that $(U \cap V) \cap Y_j \neq \emptyset$. Therefore, this implies that $U \cap V \neq \emptyset$.

Thus, Y is irreducible by the open set characterization of an irreducible set shown in class. Thus, Y is an upper bound for this chain in Σ .

Therefore, by Zorn's Lemma we have that Σ has a maximal element W s.t. $Z \subset W$ and W is an irreducible component since W is maximal. ■

(c)

Prove that X is the union of its irreducible components.

Proof.

Let $x \in X$. Consider $\{x\}$ this is trivially irreducible. Thus, by (b) we have that $\{x\} \subset W_x$ where W_x is a maximal component.

Therefore, $X \subset \bigcup_{x \in X} W_x$. Thus, $X = \bigcup_{x \in X} W_x$. ■

(d)

Let R be a ring. Prove there is a bijection

$$\{\text{minimal prime ideals } \mathfrak{p} \subset R\} \rightarrow \{\text{irreducible components of } \operatorname{Spec}(R)\}$$

given by the construction $\mathfrak{p} \rightarrow V(\mathfrak{p})$ of taking the vanishing locus of the ideal $\mathfrak{p}$.

Proof.

Define $\varphi : \{\text{minimal prime ideals } \mathfrak{p} \subset R\} \rightarrow \{\text{irreducible components of } \operatorname{Spec}(R)\}$ s.t.
 $\varphi(\mathfrak{p}) = V(\mathfrak{p})$

Well-Defined:

Let $\mathfrak{p}$ be a minimal prime. Shown in class $V(\mathfrak{p})$ since $\mathfrak{p}$ is prime this implies that $V(\mathfrak{p})$ is irreducible and closed.

Let $V(\mathfrak{p}) \subseteq Y$ s.t. Y is an irreducible closed subset. Also shown in class this implies that $Y = V(I(Y))$ s.t. $I(Y) = \mathfrak{q}$ is prime.

This implies that $V(\mathfrak{p}) \subseteq V(\mathfrak{q})$. Since $\mathfrak{p} \in V(\mathfrak{p}) \implies \mathfrak{p} \in V(\mathfrak{q})$. Thus, $\text{Problem 6} : \mathfrak{p} \subset \mathfrak{q}$.

Since $\mathfrak{p}$ is minimal this implies $\text{Problem 7} := \mathfrak{p}$.

Therefore, $V(\mathfrak{p}) = Y$. Thus, $V(\mathfrak{p})$ is an irreducible component.

Injective:

Assume $V(\mathfrak{p}) = V(\mathfrak{q})$ for $\mathfrak{p}, \mathfrak{q}$ minimal primes.

This implies that $\mathfrak{p} \subset \mathfrak{q}$ and $\mathfrak{q} \subseteq \mathfrak{p}$. Thus, $\mathfrak{q} = \mathfrak{p}$.

Surjective:

Assume Y is an irreducible component. Then $Y = V(I(Y))$ s.t. $I(Y) = \mathfrak{q}$ is prime.

If $\mathfrak{p} \subset \mathfrak{q}$, then $V(\mathfrak{q}) \subset V(\mathfrak{p})$ which implies $Y = V(\mathfrak{q}) = V(\mathfrak{p})$. Thus, $\mathfrak{q} = \mathfrak{p}$. Therefore, $\mathfrak{q}$ is minimal. Thus, $\varphi(\mathfrak{q}) = Y$.

Therefore, φ is a bijection. ■